MIL-I-37293A(DSA-DM)
26 January 1976
SUPERSEDING
MIL-I-37293
13 August 1975

MILITARY SPECIFICATION

ISOSORBIDE DINITRATE TABLETS, MODIFIED

This specification is approved for use by the Defense Personnel Support Center, Defense Supply Agency, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Isosorbide Dinitrate Tablets, Modified.

1.2 Classification. The tablets covered by this specification shall be of the following sizes (See 6.2):

Size 1 - 500s

Size 2 - Individually Sealed, 100s

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-C-186	- Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals (See 6.3).

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

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(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and
Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration. Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be Isosorbide Dinitrate Tablets, Modified and shall be in accordance with all applicable requirements of FED-STD-140, and as specified herein.

S2. Classification. Shall be type I.

S5.3 The following is added to this paragraph:

The tablets shall contain 10 mg of isosorbide dinitrate, within the designated assay limits for the tablets.

The tablets shall assay to contain not less than 90.0 percent and not more than 110.0 percent of the required amount of isosorbide dinitrate, when determined as specified in 4.3.1.

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Identification. When prepared as specified in 4.3.2, the infrared absorption spectrum (graph) of the sample shall compare qualitatively with the spectrum (graph) of Isosorbide Dinitrate Standard, as kept on file by the manufacturer. A copy of the standard infrared curve shall be forwarded to the Technical and Quality Assurance Division, Directorate of Medical Materiel, Defense Personnel Support Center, to be kept on file for use if samples are submitted for testing.

S6.4.9 Disintegration. Shall not exceed 30 minutes when determined by the U.S.P. method for uncoated tablets, using Simulated Gastric Fluid T.S. (U.S.P.) as the immersion fluid.

3.2 Expiration dating period. For size 1 the expiration dating period shall be 60 months, and for size 2 the expiration dating period shall be not less than 24 months.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period (potency period) specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the items to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Sampling. The sampling as shown in FED-STD-140 shall be applicable.

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4.3 Tests

4.3.1 Assay.

Reagents:

Phenoldisulfonic Acid. Heat 5 Grams of colorless phenol with a mixture of 30 ml concentrated H_2SO_4 and 15 ml fuming H_2SO_4 (containing approximately 20 percent free SO_3) on a steam bath for 2 hours. Store in an amber, glass stoppered bottle.

14 percent Ammonia Water. Mix equal volumes of concentrated ammonia with purified water.

Procedure:

Weight and finely powder 20 tablets. Quantitatively transfer an accurately weighed sample of the powdered tablets, equivalent to 20 mg of isosorbide dinitrate (2 tablets), into a 125 ml glass-stoppered Erlenmeyer flask. Place in the flask a Teflon-coated magnetic stirring bar and pipet into the flask exactly 50 ml of glacial acetic acid. Stir magnetically for 30 minutes: Filter through a No. 12 Whatman filter paper.

Prepare a standard stock solution by dissolving 100 mg sodium nitrate standard; previously dried at 105°C, in 1 ml of purified water in a 100 ml volumetric flask. Dilute to volume with glacial acetic acid, and mix well. Prepare a working standard by diluting exactly 14.40 ml of the standard stock solution to exactly 50 ml in a volumetric flask, using glacial acetic acid (equivalent to 20 mg of isosorbide dinitrate); mix well. To the first of three separate 50 ml volumetric flasks add 1 ml of glacial acetic acid, to the second flask add 1 ml of the final diluted standard solution, and to the third flask add 1 ml of the sample solution. To each flask add 2 ml phenoldisulfonic acid reagent, mix well and allow to stand for 15 minutes. To each flask add 20 ml purified water and 20 ml ammonia water (14 percent NH_3), mix and cool to room temperature. Dilute the contents of each flask to volume with purified water, and mix well. Determine the absorbance of each solution at 408 nm with a suitable spectrophotometer, using the blank solution (first flask) in the reference cell.

As an alternate, the assay method for Isosorbide Dinitrate Tablets in the USP shall be acceptable.

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4.3.2 Identification.

Weigh an appropriate amount of finely powdered tablet powder into a 35-ml glass-stoppered centrifuge tube. Add 10 ml of 0.1N NaOH and shake well until all of the tablet ingredients are wetted. Immediately add 10 ml of petroleum ether and shake well. Centrifuge and remove, by means of an eye dropper, 50 to 70 percent of the petroleum ether. Transfer to an agate mortar and evaporate to dryness. Prepare a pellet by incorporating the residue into approximately 200 mg of KBr. Record the infrared spectrum form 2 to 5 μ.

As an alternate, the identification for Isosorbide Dinitrate Tablets in the USP shall be acceptable.

4.4 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in table I. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

TABLE I. Examination of preparation for delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and Packing Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper location, sequence, or method of application.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable paragraphs of PPP-C-186 and as specified herein.

5.2 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.2.1 Unit of issue. One bottle containing five hundred tablets (size 1) or one box containing one hundred individually sealed tablets (size 2), as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container.

5.2.2.1.1 Size 1. For size 1 the immediate container shall be a commercially available bottle considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.1.2 Size 2. For size 2 the immediate container shall be a commercially available blister or strip packaging considered acceptable to the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2 Unit package.

5.2.2.2.1 Size 1. At the option of the contractor, a unit package may be furnished for each bottle containing size 1 tablets. If a unit package is furnished, each bottle shall be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.2.2 Size 2. One hundred immediate containers containing size 2 tablets shall be packaged in a box conforming to PPP-B-566 or PPP-B-676. Closure shall be as specified in the appendix of the box specification.

5.2.2.3 Intermediate package. Twelve unit packages containing size 1 tablets shall be packaged as specified in 5.2.8 of PPP-C-186. An intermediate package is not required for size 2 tablets.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.3 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.3.1 Level A. Four intermediate packages (48 units) of size 1 tablets or ten unit packages of size 2 tablets shall be packaged in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

5.3.2 Level B. four intermediate packages (48 units) of size 1 tablets or ten unit packages containing size 2 tablets shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the expiration date.

5.4.1.2 Unit package. Each unit package label shall bear the following:

(a) labeling information in accordance with commercial practice.
(b) the National Stock Number.

NOTES: (1) The National Stock Number shall not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Labeling adhesion test shall not apply.
(4) If a unit package is not supplied for size 1, the National Stock Number shall appear on the immediate container.

5.4.2 Intermediate package. Each intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. Type II Shelf Life marking as specified in MIL-STD-129 shall be shown for size 1 tablets and Type I Shelf Life markings shall be shown for size 2 tablets. Markings shall include the legend:

"STORE AT CONTROLLED ROOM TEMPERATURE (59° - 86° F.)".

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5.4.3 Exterior container. Exterior containers shall be marked as specified in 5.5.4 of PPP-C-186. Type II Shelf Life marking as specified in MIL-STD-129 shall be shown for Size 1 tablets and Type I Shelf Life markings shall be shown for size 2 tablets. Marking shall include the legend:

"STORE AT CONTROLLED ROOM TEMPERATURE (59° - 86° F.)"

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) the National Stock Number.
(c) Selection of applicable levels of packaging and packing (See 5.2 and 5.3).

6.2 This specification covers the following item(s):

National Stock No.	Item Identification	Specification Designation
6505-00-761-1506	ISOSORBIDE DINITRATE TABLETS, MODIFIED, 10 mg, 500s	Size 1
6505-01-013-1303	ISOSORBIDE DINITRATE TABLETS, MODIFIED, 10 mg, Individually Sealed, 100s	Size 2

6.3 All references to PPP-C-186 refer to Interim Federal Specification PPP-C-00186B which is mandatory for use until superseded by a revision.

6.4 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1612

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-I-37293A(DSA-DM) ISOSORBIDE DINITRATE TABLETS, MODIFIED

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

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